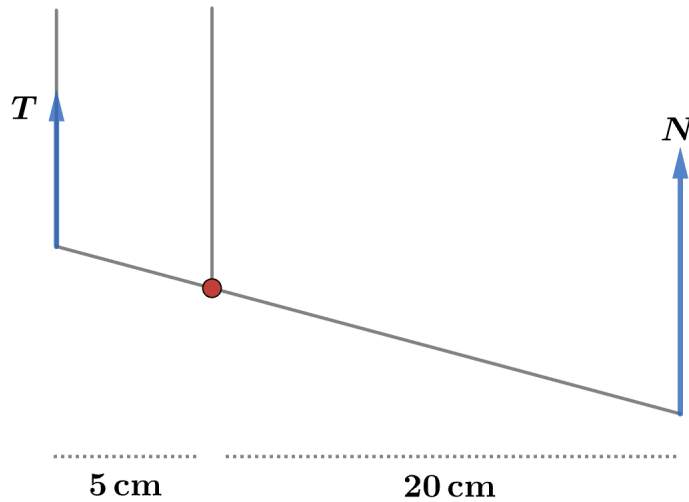


2013 F=ma Exam: Problem 22

Kevin S. Huang



Choosing the ankle joint as the pivot point, only the normal force from the ground and tension in the Achilles tendon contribute to the torque:

$$T(5\text{ cm}) = N(20\text{ cm})$$

Since the student is standing on a single toe, $N = mg$ so we have

$$T = 4N = 4mg = 4(60\text{ kg})(10\text{ m/s}^2) = 2400\text{ N}$$

so the answer is D.